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Compare Linear and Polynomial Regression using Python

CODE :

```
import numpy as np

X = np.array([1, 2, 3, 4, 5, 6])
Y = np.array([2, 5, 10, 17, 26, 37])

x_mean = np.mean(X)
y_mean = np.mean(Y)

m = np.sum((X - x_mean) * (Y - y_mean)) / np.sum((X - x_mean) ** 2)
c = y_mean - m * x_mean

linear_pred = m * X + c

# Fit polynomial:  $y = ax^2 + bx + c$ 
poly_coeff = np.polyfit(X, Y, 2)
polynomial_pred = np.polyval(poly_coeff, X)

linear_mse = np.mean((Y - linear_pred) ** 2)
polynomial_mse = np.mean((Y - polynomial_pred) ** 2)

print("===== LINEAR REGRESSION =====")
print("Equation: Y =", round(m, 2), "X +", round(c, 2))
print("Predicted Values:", np.round(linear_pred, 2))
print("MSE:", round(linear_mse, 2))

print("\n===== POLYNOMIAL REGRESSION =====")
print(
    "Equation: Y =",
    round(poly_coeff[0], 2), "X2 +",
    round(poly_coeff[1], 2), "X +",
    round(poly_coeff[2], 2)
)
print("Predicted Values:", np.round(polynomial_pred, 2))
print("MSE:", round(polynomial_mse, 2))
```

```
print("\n===== COMPARISON =====")

if polynomial_mse < linear_mse:
    print("Polynomial Regression performs better.")
elif linear_mse < polynomial_mse:
    print("Linear Regression performs better.")
else:
    print("Both models have the same performance.")
```

OUTPUT

```
===== LINEAR REGRESSION =====
Equation: Y = 7.0 X + -8.33
Predicted Values: [-1.33  5.67 12.67 19.67 26.67 33.67]
MSE: 6.22

===== POLYNOMIAL REGRESSION =====
Equation: Y = 1.0 X2 + -0.0 X + 1.0
Predicted Values: [ 2.  5. 10. 17. 26. 37.]
MSE: 0.0

===== COMPARISON =====
Polynomial Regression performs better.

=== Code Execution Successful ===
```